

Modeling Mental Health Treatment-Seeking Behavior to Inform Workplace Support Strategies

Presented by Adil Alimohd and Tania Amreen

DSC 510 - Health Data Science

Instructor - Casey Bennett

Introduction

- Mental health is a population health crisis — **1 in 5 U.S. adults are affected**, and 50% will experience a disorder in their lifetime
- Employers face **\$6 trillion in projected productivity losses** due to mental health challenges
- Psychologically healthy workplaces lead to measurable improvements in employee well-being
- Despite this, many employees who need support **do not seek treatment** due to stigma and lack of awareness
- This project applies the CDC HI-5 model across three phases — classification and clustering to identify and segment at-risk employee populations, a community-based intervention to reach them through targeted outreach, and a data engineering infrastructure to operationalize and sustain it

Project Approach

- Goal: identify employees who need support but do not seek treatment
- Use **classification** to predict treatment-seeking behavior
- Use **clustering** to segment employees based on shared barriers
- Design **targeted interventions** for each segment
- Develop a **data pipeline** to operationalize and scale the solution

Dataset Description

Dataset Overview

- Source: Kaggle workplace mental health survey
- 1,259 respondents, 27 features (demographics, work environment, mental health perceptions)

Target Variable

- Treatment (Yes/No), ~balanced (51% / 49%) → suitable for binary classification

Data Cleaning

- Removed irrelevant columns
- Standardized gender categories
- Filtered ages (18–75)
- Imputed missing values

Final Dataset

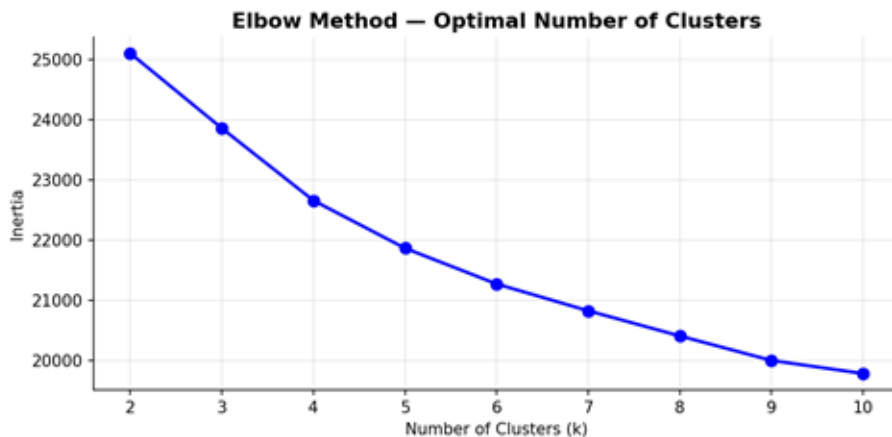
- 1,251 rows × 24 columns, no missing values

Use

- Population-level analysis of workforce mental health and barriers to care
- Aligned with CDC HI-5 Competency #1

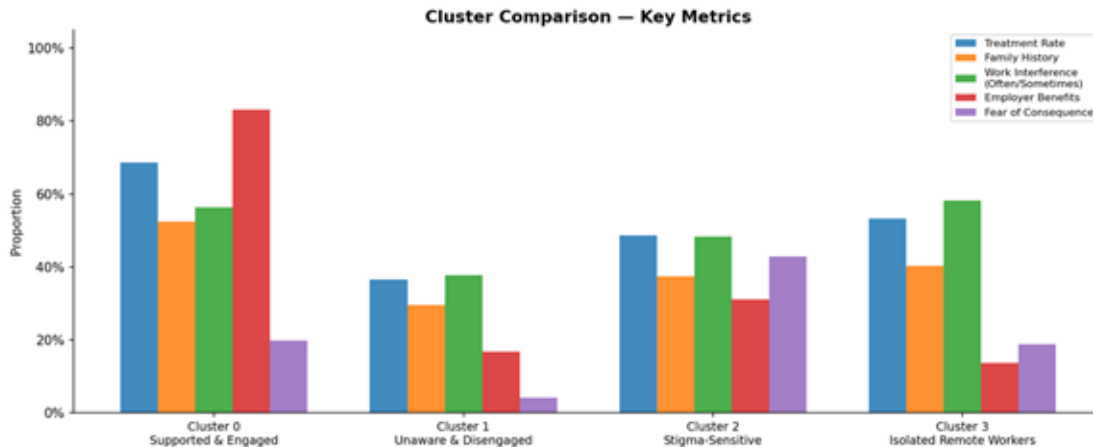
Clustering Methodology

- K-Means clustering was applied to the full cleaned dataset to identify natural groupings of employees based on their demographics, work environment, and mental health attitudes.
- All categorical features were encoded and scaled prior to clustering.
- The elbow method was used to evaluate $k=2$ through $k=10$, with the curve flattening around $k=4$, which was selected as the optimal number of clusters.



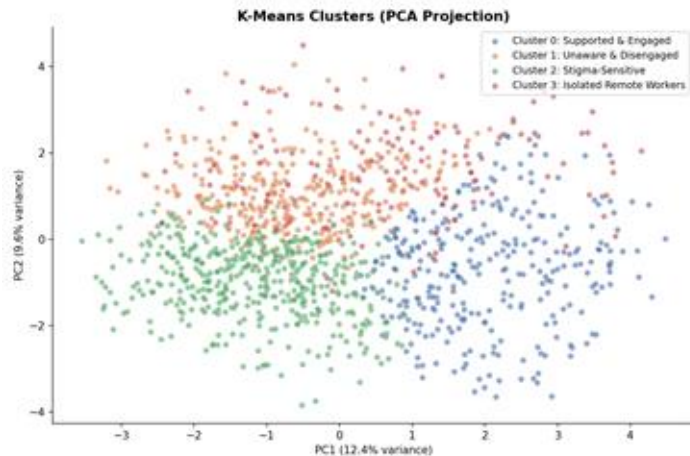
Cluster Profiles

- Four distinct employee populations emerged from the analysis, each characterized by different barriers to seeking mental health support.
 - Cluster 0 (Supported & Engaged, n=309) has strong employer support and a 69% treatment rate.
 - Cluster 1 (Unaware & Disengaged, n=369) has minimal employer benefits and the lowest treatment rate at 37%.
 - Cluster 2 (Stigma-Sensitive, n=434) reports the highest fear of professional consequences at 43% with very low anonymity protections.
 - Cluster 3 (Isolated Remote Workers, n=139) is 71% remote with minimal employer support and high work interference.



Cluster Visualization (PCA)

- To visualize the clusters, Principal Component Analysis was used to reduce the dataset from 24 dimensions down to 2, allowing the cluster groupings to be plotted.
- PC1 captures 12.4% of total variance and PC2 captures 9.6%, meaning the two components together explain roughly 22% of the total variance in the data.
- The overlap between clusters in 2D is expected given the high dimensionality of the original dataset. The clusters are meaningfully distinct in the full 24-dimensional feature space even if they appear to blend in this projection.
- The separation visible between Cluster 0 (blue, right) and Cluster 2 (green, left) along PC1 reflects the strong contrast in employer support and anonymity protections between the two groups.



Clustering Results & Future Directions

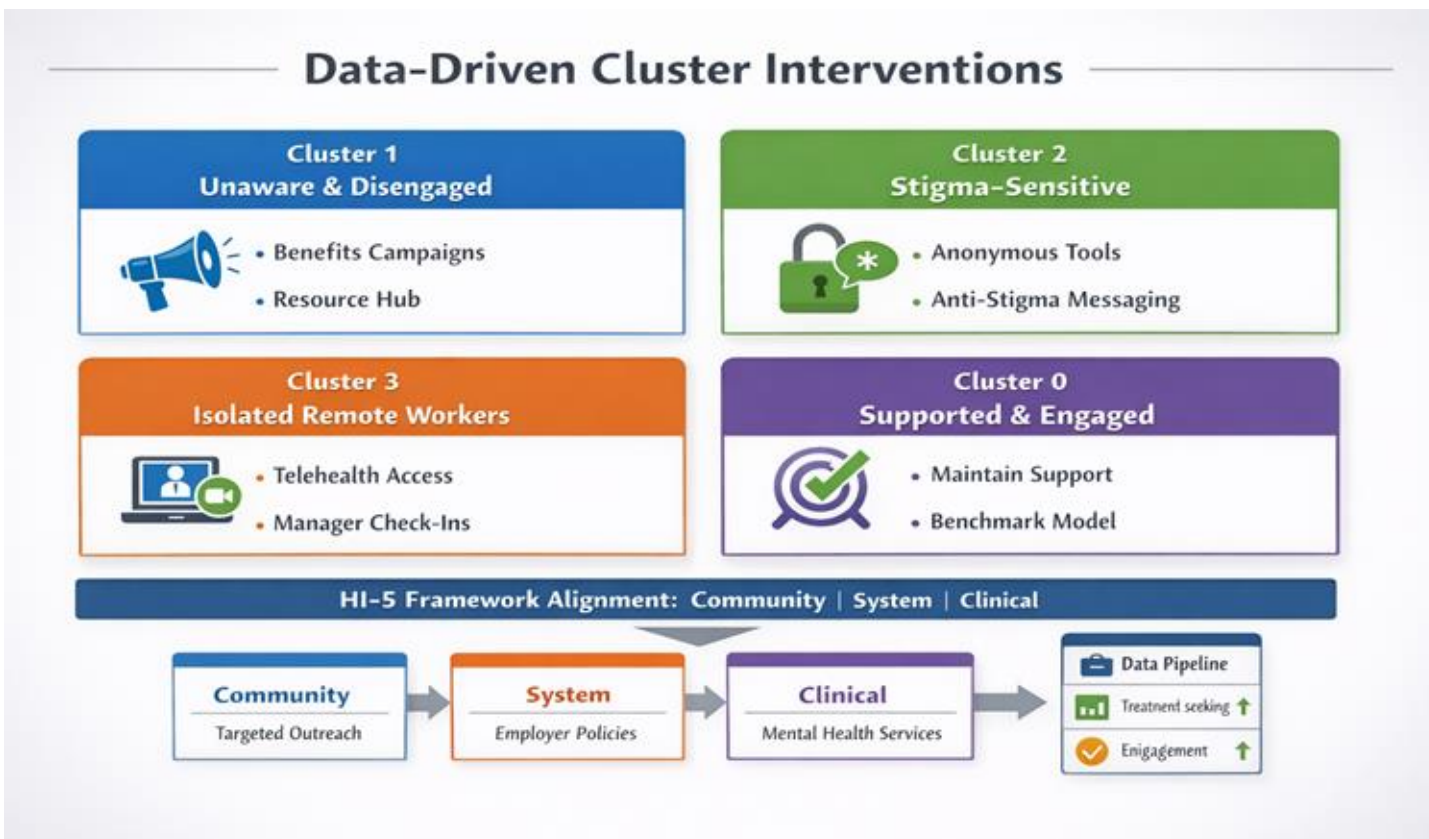
Analysis Results:

- K-Means identified 4 distinct employee segments based on demographics, workplace conditions, and mental health attitudes
- Each cluster highlights a different barrier to care, with measurable differences in treatment-seeking behavior and employer support
- PCA visualization confirms meaningful separation, while overlap reflects high-dimensional structure rather than poor clustering

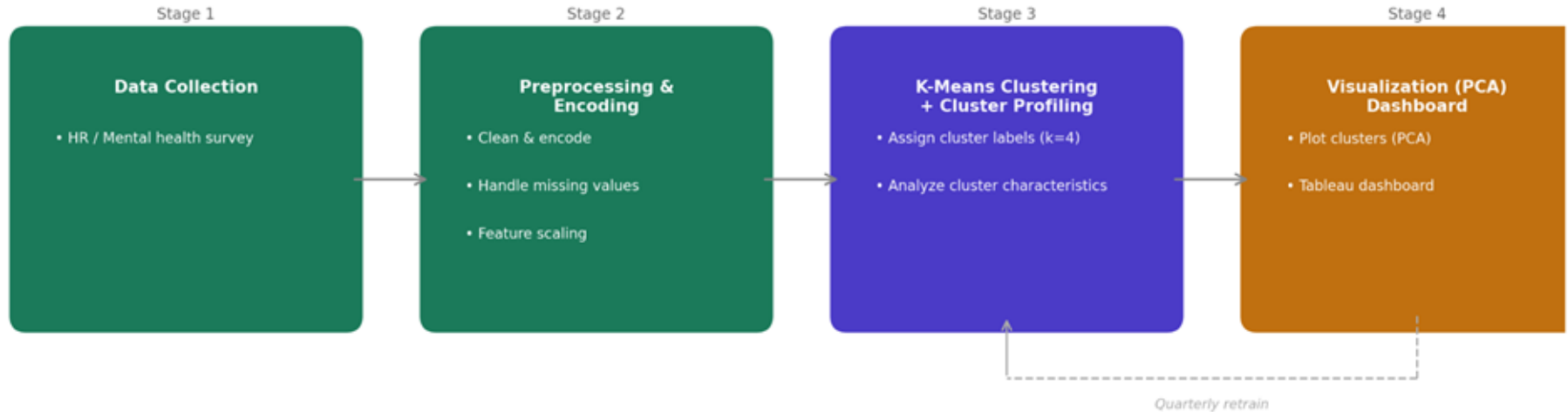
Future Improvements:

- Validate cluster stability using alternative methods (e.g., silhouette score, hierarchical clustering)
- Incorporate additional features (e.g., tenure, job role, organizational culture indicators) to refine segmentation
- Track clusters over time to detect shifts in employee populations
- Re-run clustering periodically to capture evolving workplace conditions and improve intervention targeting
- Link cluster membership to downstream outcomes to better quantify impact of interventions

Clustering Intervention



Data Engineering - Pipeline



Classification Methodology

- Target Variable - Employees who had a mental health condition but did not seek help
 - Feature Engineering
- 4 different models tested
 - Logistic Regression
 - Decision Trees
 - Random Forest
 - Gradient Boosting

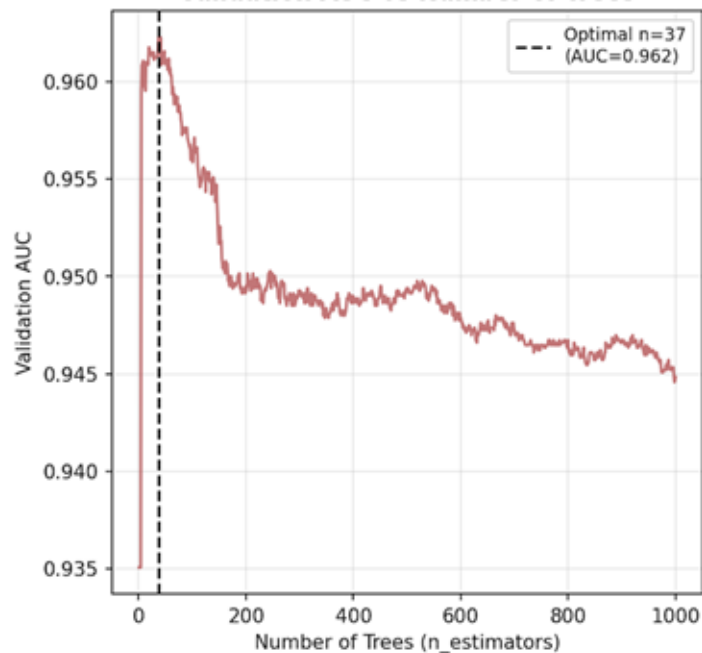
Classification Target Variable Selection

- If you have a mental health condition, do you feel that it interferes with your work?
- Does your employer provide resources to learn more about mental health issues and how to seek help?

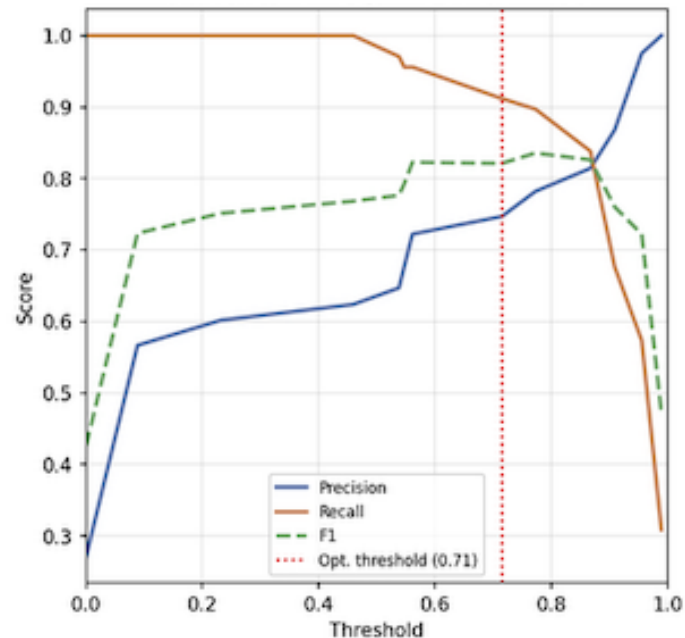
work_interfere	Total	seek_help = N	Rate
Never	477	227	47.6%
Rarely	173	79	45.7%
Sometimes	465	247	53.1%
Often	144	93	64.6%

Classification - Performance Tuning

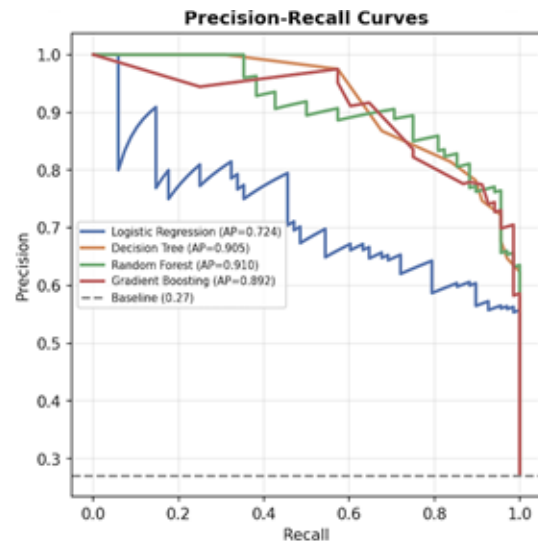
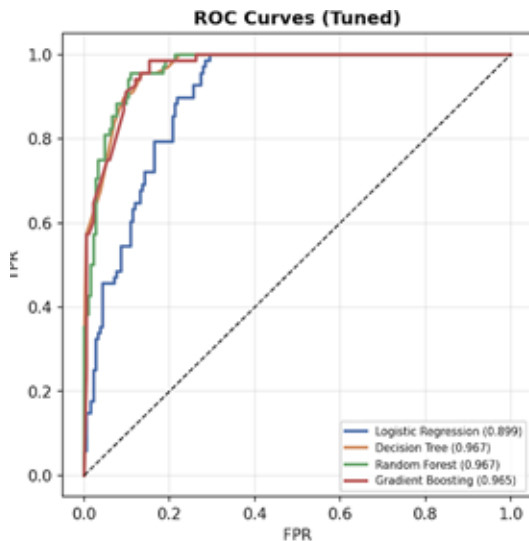
Early Stopping — Gradient Boosting
Validation AUC vs Number of Trees



Threshold Sweep



Classification Results - Model Output Comparisons



Model	CV AUC	Test AUC	Avg Precision	Opt. Recall	Opt. Precision	Employees Flagged
Logistic Regression	0.896	0.899	0.724	0.926	0.573	110
Decision Tree	0.946	0.967	0.905	0.912	0.747	83
Random Forest	0.957	0.967	0.91	0.941	0.771	83
Gradient Boosting	0.958	0.965	0.892	0.912	0.775	80

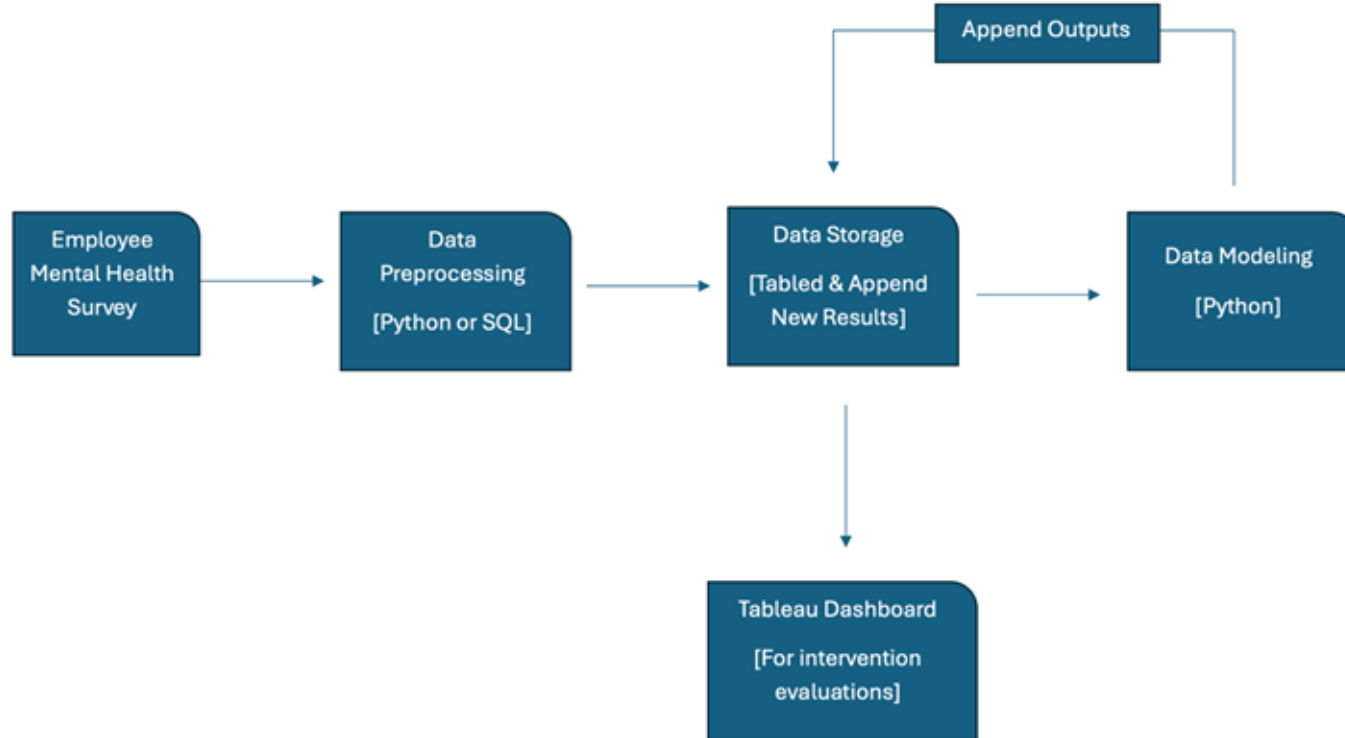
Classification Intervention - Benefits

- Recommendation #1: Expand Benefit Offerings
 - Digital Health Vendors - Lyra Health & Spring Health
 - Paid Time Off - Mental Health Days
 - Employee Assistance Programs (EAPs)
 - Assessments
 - Short Term Counseling
 - Referrals

Classification Intervention - Leadership Trainings

- Recommendation #2: Manager/Leadership Trainings
 - Attempt to influence the environment
 - Required Trainings focused on
 - Recognition & Awareness
 - Communication
 - Crisis Response
 - Reducing Stigma
 - Environment Building

Classification Intervention - Data Pipeline



Further Analysis Opportunities

- Evaluation of Intervention Success
 - A/B Testing
- Root Cause Analysis
 - Would require additional data (unused survey fields, leave records, SDOH)
- Developing Risk Scores to further segment the population and track fluctuations

Considerations

- Company Costs
 - Implementation
 - Survey
 - Maintenance of data pipeline/dashboards
- Survey Frequency
 - What is the correct cadence to maintain participation
 - Incentives?
- Privacy
 - Big Brother?

Conclusion

- Mental health improvements in a working population can be advanced through classification and clustering methods applied onto a population's survey data
- Random Forest Classification
 - 83 identified employees
- K-Means Clustering
 - 4 clustered groups - Supported & Engaged / Unaware & Disengaged / Stigma-Sensitive / Isolated Remote Workers
- Interventions of Benefit Offering Expansions, Leadership Trainings, and Targeted Communications